

CO4 Assessment Tool 1 – Solutions

Solutions for Questions 9, 20, 22, 25 and 26

Question 9 – YOLO Performance across Scales

The given detection results are:

Object Size	Detection Accuracy
Large	95%
Medium	85%
Small	60%

Analysis

YOLO performs very well for large objects, with 95% accuracy. Its performance decreases to 85% for medium objects and drops considerably to 60% for small objects. The difference between large- and small-object accuracy is $95\% - 60\% = 35$ percentage points. This shows that object scale has a significant effect on detection performance.

Conclusion

YOLO alone may not be sufficient when reliable detection is required across all object scales, especially for small objects. Improvements can include multi-scale feature extraction, higher input resolution, feature-pyramid or feature-fusion methods, suitable data augmentation, and training with more small-object examples.

Question 20 – Surveillance Model Selection

The given surveillance model results are:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Analysis

Under strict latency constraints, Model A has the lowest latency at 30 ms. Model B provides higher accuracy of 90% but has a latency of 70 ms. Model C has the highest accuracy of 95%, but its latency of 120 ms makes it unsuitable when response time is critical.

Conclusion

Model A should be selected under strict latency constraints because it provides the fastest response. If the system can tolerate a moderate increase in latency, Model B is a better accuracy-latency compromise.

Question 22 – Depth Estimation Error Analysis

The given depth-estimation errors are:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Analysis

For depth estimation, a lower error indicates better accuracy. Method A has the lowest error of 0.5 m, followed by Method C at 0.8 m. Method B has the highest error at 1.2 m. Therefore, the accuracy ranking is A, C, then B.

Conclusion

Method A is the most accurate and is the preferred choice for real-world applications where accurate depth measurement is important. Method C is a possible alternative, while Method B is the least suitable because it has the largest error.

Question 25 – Model Efficiency vs Accuracy

The given model efficiency and accuracy results are:

Model	Accuracy	Computation
A	90%	High
B	85%	Low
C	88%	Moderate

Analysis

For edge devices, both accuracy and computational efficiency are important because resources are limited. Model A has the highest accuracy at 90% but requires high computation. Model B has low computation but lower accuracy of 85%. Model C provides 88% accuracy with moderate computation, giving a balanced trade-off.

Conclusion

Model C is the best choice for edge devices because it provides good accuracy while requiring only moderate computation. Model A can be chosen when maximum accuracy is more important and sufficient hardware resources are available.

Question 26 – Feature Detection under Illumination Changes

The given feature-detection results are:

Method	Normal Light	Low Light
A	90%	60%
B	85%	75%

Analysis

Method A performs better under normal illumination, with 90% accuracy, but its accuracy drops to 60% under low light. Method B has 85% accuracy under normal light and 75% under low light. The accuracy drop is 30 percentage points for A and only 10 percentage points for B. Therefore, Method B is more robust to illumination changes.

Conclusion

Method B is preferable for applications where lighting conditions vary. Although Method A performs better in normal lighting, its large degradation in low-light conditions makes it less reliable.

Programming Requirement

None of Questions 9, 20, 22, 25, or 26 requires a program according to the uploaded assessment document. Therefore, input/output screenshots are not applicable to these five questions.